Bibliography and References Cited

[1] S. Roy, J. R. Gord, A. K. Patnaik, "Recent advances in coherent anti-Stokes Raman scattering spectroscopy: Fundamental developments and applications in reacting flows," Prog. Energy Combust. Sci. 36 (2010) 280–306.

[2] J. D. Miller, M. N. Slipchenko, T. R. Meyer, H. U. Stauffer, J. R. Gord, "Hybrid femtosecond/picosecond coherent anti-Stokes Raman scattering for high-speed gas-phase thermometry," Opt. Lett. 35 (2010) 2430–2432, doi:10.1364/OL.35.002430.

[3] S. P. Kearney, "Hybrid fs/ps rotational CARS temperature and oxygen measurements in the product gases of canonical flat flames," Combust. Flame 162 (2015); Sandia report SAND2014-17275J, OSTI 1183118.

[4] A. M. Weiner, "Ultrafast optical pulse shaping: A tutorial review," Opt. Commun. 284 (2011) 3669–3692.

[5] O. Busari, "High-Repetition-Rate fs CARS: Instrument Basis," technical memorandum (physics, engineering and calculation basis, including the density-matrix cross-check), Morgan State University, October 2026, unpublished; available on request.

[6] A. C. Eckbreth, *Laser Diagnostics for Combustion Temperature and Species*, 2nd ed., Gordon and Breach, Amsterdam, 1996.

[7] R. C. Millikan, D. R. White, "Systematics of vibrational relaxation," J. Chem. Phys. 39 (1963) 3209–3213.

[8] N. Rodrigues, O. Busari, W. Senior, C. McDonald, Y. Chen, A. North, W. Laster, S. Meyer, R. Lucht, "NOx reduction in an axially staged gas turbine model combustor through increase in the combustor exit Mach number," Combust. Flame 212 (2020) 282–294, doi:10.1016/j.combustflame.2019.10.039.

[9] Office of Naval Research, Code 352, Hypersonic Aerothermodynamics, High-Speed Propulsion and Materials, research topics, https://www.onr.navy.mil/organization/departments/code-35/division-352/hypersonic-aerodynamics (accessed September 2026).

[10] V. Anand, E. Gutmark, "Rotating detonation combustors and their similarities to rocket instabilities," Prog. Energy Combust. Sci. 73 (2019) 182–234.

[11] U.S. Standard Atmosphere, 1976, NOAA-S/T 76-1562, NOAA, NASA and U.S. Air Force, Washington, DC, 1976.

[12] O. Busari, uncertainty analysis for the sub-atmospheric Hencken reference flame (±16 K at 50 Torr, ±41 K at 760 Torr), Morgan State University, 2026, unpublished.

[13] P. Panda, O. Busari, M. Roa, R. Lucht, "Flame stabilization mechanism in reacting jets in swirling vitiated crossflow," Combust. Flame 207 (2019) 302–313, doi:10.1016/j.combustflame.2019.06.005.

[14] P. Panda, O. Busari, R. Lucht, W. Laster, "Effect of the nature of vitiated crossflow on the flow-field of a transverse reacting jet," Exp. Fluids 58(2) (2017), doi:10.1007/s00348-016-2288-4.

[15] N. Rodrigues, C. McDonald, O. Busari, A. Satija, A. North, W. Laster, S. Meyer, R. Lucht, "An optically accessible secondary combustion zone for the transverse injection of reacting jets into a high-speed, vitiated crossflow within a staged, gas turbine model combustor," Meas. Sci. Technol. 32(2) (2020) 024007, doi:10.1088/1361-6501/abbd57.

[16] N. Rodrigues, C. McDonald, O. Busari, A. Satija, R. Lucht, "Transverse injection of rich, premixed, natural gas–air and natural gas–hydrogen–air reacting jets into high-speed vitiated crossflow at engine-relevant conditions," Int. J. Hydrogen Energy 46(72) (2021) 35718–35738, doi:10.1016/j.ijhydene.2021.08.108.